



FYDP Proposal Form

Department of Computer Science

Batch 2022-2026

FYDP Title

DevDialogue AI - Requirement-Aware AI Code Generation System

Domain

Artificial Intelligence / Web Development

Nature of FYDP

- ☐ New Project
 ☐ Industrial Collaboration
 ☐ Funded Project
☒ Extension of Existing Project
 ☐ Other Department / Academic Institution

Problem Statement

Current AI-powered code generation platforms (e.g., ChatGPT, Gemini, Grok, DeepSeek) suffer from a fundamental limitation: they generate code based on vague or incomplete user prompts, producing output that is either over-engineered or missing key requirements. As a result, developers enter a repetitive trial-and-error cycle of copying code into their editor, identifying errors, returning to the AI with corrective prompts, and repeating the process. This consumes significant development time, increases cognitive load, and reduces trust in AI tools. The core problem is the absence of a structured, dialogue-driven mechanism that collects complete project requirements before generating code.

Proposed Solution (FYDP)

DevDialogue AI is a web-based platform that addresses the prompt-quality gap through two complementary modes. Mode 1 (Dialogue-Driven): The user submits a basic project description, and the system engages in a structured conversational interview, asking targeted clarifying questions about modules, data flow, expected behavior, and edge cases. Once sufficient context is gathered, the AI generates a complete, fully-functional codebase. Mode 2 (Incremental Generation): The system generates code step-by-step, presenting each module individually, asking questions after each step, and progressively building the full project based on user responses and suggestions. The platform uses a dual-layer architecture: Django MVT serves as the main backend handling user authentication, session management, and the web interface, while N8N (workflow automation) orchestrates the core AI logic — managing the dialogue/question flow between the user and the AI, and automating DeepSeek API calls within each workflow stage. This separation allows the AI workflow to be visually designed, modified, and scaled independently of the web layer.

Scope of the Project

The scope of this project is limited to software/code generation tasks only. The platform will support web-based interaction via a browser interface and will not include IDE plugins or offline usage. The system will handle Python-based and web development projects as primary use cases. The platform will manage user sessions, store conversation history, and support multi-step requirement elicitation. It will not cover hardware programming, mobile native app development, or real-time collaborative editing. The backend will be powered by Django with SQLite as the database, and code generation will rely exclusively on the DeepSeek API.

Development / Research Methodology

The project follows an Agile development methodology with iterative sprints. Phase 1 involves requirements analysis and system design, including user journey mapping, API integration planning, and N8N workflow design. Phase 2 covers Django MVT backend development including user authentication and session management. Phase 3 focuses on N8N workflow setup — building automated pipelines for DeepSeek API calls and designing the dialogue/question-generation flow for both modes. Phase 4 involves connecting Django with N8N via webhooks, enabling the frontend to trigger N8N workflows and receive generated code responses. Phase 5 covers frontend development using Django templates. Phase 6 covers testing (unit, integration, and user acceptance testing). Phase 7 is deployment and documentation. Research methodology includes comparative analysis of existing AI code generation tools and iterative user testing to refine the dialogue strategy.

Resources Involved

Backend Framework: Django (Python) - MVT architecture.
Workflow Automation: N8N (self-hosted) - for orchestrating DeepSeek API calls and dialogue flow automation.
Programming Language: Python 3.x.
AI API: DeepSeek API for code generation (called via N8N workflows).
Database: SQLite (db.sqlite3 - Django default).
Frontend: Django HTML Templates, CSS, JavaScript.
Integration: Django-N8N communication via Webhooks.
Version Control: Git / GitHub. Development Tools: VS Code, Postman (API testing). Hosting: Local development server (potential cloud deployment).
Hardware: Standard development laptops/PCs for each team member.

Final Outcome / Deliverables

1. A fully functional web application (DevDialogue AI) supporting two code generation modes.
2. A dialogue engine capable of generating context-aware clarifying questions based on user prompts.
3. Integration with DeepSeek API for intelligent, requirement-complete code generation.
4. User session management with conversation history.
5. A tested and documented Django-based backend system.
6. Final project report and technical documentation.
7. Demonstration video and presentation slides.

Learning Outcome

Upon completion of this project, students will be able to:

- (1) Design and implement a full-stack web application using Django MVT architecture.
- (2) Integrate and orchestrate third-party AI APIs (DeepSeek) using N8N workflow automation.
- (3) Apply software engineering principles including requirements elicitation, system design, and iterative development.
- (4) Build automated AI dialogue workflows using N8N for context-aware question generation and code delivery.
- (5) Understand the challenges of human-AI interaction in software development tools.
- (6) Apply database management using SQLite within Django ORM.
- (7) Gain hands-on experience with low-code workflow automation tools (N8N) in real-world AI system integration.
- (8) Gain practical experience in Agile project management and team collaboration.

Industrial Support (if any) _____

FYDP Team members:

S. No.	Name of the Student	IU Reg. No	CGPA	Signature
1	Syed Wafa Abbas Naqvi	37249	2.38	
2	Shehryar Mobashar	37189	2.42	
3	Shehryar Khan	37045	2.95	

Note: Group cannot be more than 4 students

Details of Supervisor / Industrial Advisor:

	Name	Designation & Department	Contact No.	Signature
Supervisor	Sumaira Aslam			
Co-Supervisor (If any)				
Industrial Advisor				

Project Schedule

Attached Project schedule with your FYDP Proposal using MS-Project (or similar tools) with all tasks, deliverables, milestones, cost estimates, clearly indicated.

FYDP to Sustainable Development Goals (SDGs) Mapping:

SDGs	Mapping	Description
SDG 01. No poverty		
SDG 02. Zero hunger		
SDG 03. Good health and well-being		
SDG 04. Quality education	✓	SDG 04 - Quality Education: DevDialogue AI improves coding education by teaching users structured requirements thinking and AI-assisted development.
SDG 05. Gender equality		
SDG 06. Clean water and sanitation		
SDG 07. Affordable and clean energy		
SDG 08. Decent work and economic growth	✓	SDG 08 - Decent Work and Economic Growth: The platform improves developer productivity and reduces time wasted in iterative coding, supporting efficient digital work.
SDG 09. Industry, innovation, and infrastructure	✓	SDG 09 - Industry, Innovation and Infrastructure: DevDialogue AI contributes to innovation in software development tooling through AI-driven requirement elicitation.
SDG 10. Reduced inequalities		
SDG 11. Sustainable cities and communities		

SDG 12. Responsible consumption and production		
SDG 13. Climate action		
SDG 14. Life below water		
SDG 15. Life on land		
SDG 16. Peace, justice and strong institutions		
SDG 17. Partnerships for the goals	√	SDG 17 - Partnerships for the Goals: The system integrates third-party AI API (DeepSeek), demonstrating cross-platform collaboration and open technology partnerships.

Note: Marked those mapped

FYDP to Complex Computing Problem (CCP) Mapping:

CCP Attribute	WP1 and some or all WP2 to WP7:	FYDP to CCP Mapping How it is addressed in FYDP
Depth of knowledge required	WP1: one or more of ACM Knowledge Ara	The project requires deep knowledge across AI/ML, Software Engineering (SE), Human-Computer Interaction (HCI), Platform-based Development (PBD), and workflow automation (N8N). It involves integrating LLM APIs via automated N8N pipelines and designing intelligent dialogue systems well beyond standard coursework.
Range of conflicting requirements	WP2: wide-ranging or Conflicting technical, engineering/computing and other issues	Requirements conflict between usability (simple interface), completeness (thorough questioning), and performance (fast AI responses). The system must balance asking enough questions without overwhelming the user.
Depth of analysis required	WP3: no obvious solution	There is no single obvious solution to the requirement-elicitation problem for AI code generation. The team must research and design a novel dialogue-driven methodology combining conversational AI with structured requirements engineering.
Familiarity of issues	WP4: Involve infrequently encountered issues	The problem of AI code generation drift and requirement misalignment is a relatively new challenge in the field of AI-assisted development, not yet covered by standard tools or well-established practices.
Extent of applicable codes	WP5: outside problems encompassed by standards and codes of practice	The project operates at the boundary of software engineering standards and AI ethics. There are no existing codes of practice for requirement-aware AI code generation systems, requiring the team to define their own quality and interaction standards.
Extent of stakeholder involvement and conflicting requirements	WP6: diverse groups of stakeholders with widely varying needs	Stakeholders include student developers, professional developers, educators, and AI tool vendors. Each group has different expectations regarding code quality, interactivity, and learning outcomes, requiring the system to accommodate diverse needs.
Interdependence	WP7: many component parts or sub-problems	The system consists of multiple interdependent sub-problems: dialogue engine, question generation, DeepSeek API integration, code assembly, session management, and frontend interaction, all tightly coupled and requiring coordinated development.

For office us only:

☐ Proposal Approved	☐ Not Approved	☐ Returned for Clarification / Modification	
Comments (If any):			
Project Serial No.		Date:	
Signature FYDP Coordinator		Signature FYDP Convener	
Signature Associate Dean / Dean			

Annexure – A

Knowledge Areas in ACM

S#	ACM Knowledge Area	Your Project Mapping to ACM Knowledge Area
1	AL-Algorithms and Complexity	Partially applicable. Code generation and question sequencing involve algorithmic logic for determining relevant follow-up questions based on user input.
2	AR-Architecture and Organization	
3	CN-Computational Science	
4	DS-Discrete Structures	
5	GV-Graphics and Visualization	
6	HCI-Human-Computer Interaction	Directly applicable. The platform's dialogue interface requires designing user-centric interactions for collecting requirements and presenting code. Conversational UI is a core HCI challenge.
7	IAS-Information Assurance & Security	
8	IM-Information Management	
9	IS-Intelligent Systems	
10	NC-Networking and Communication	
11	OS-Operating Systems	
12	PBD-Platform-based Development	Highly applicable. The project is built on Django as the primary web platform, with N8N as a secondary automation platform. Both are platform-based tools,

		making PBD the core development paradigm across the entire system architecture.
13	PD-Parallel and Distributed Computing	
14	PL-Programming Languages	
15	SDF-Software Development Fundamentals	
16	SE-Software Engineering	Highly applicable. The project applies SE methodologies including requirements engineering, system design, testing, and Agile development throughout the lifecycle.
17	SF-Systems Fundamentals	
18	SP-Social Issues and Professional Practice	